```python
from collections import deque

def water_jug():
    q = deque([(0, 0)])
    visited = set()

    while q:
        a, b = q.popleft()

        if (a, b) in visited:
            continue
        visited.add((a, b))

        print(a, b)

        if a == 2:
            print("2 gallons obtained!")
            return

        states = [
            (4, b), (a, 3), (0, b), (a, 0),
            (a - min(a, 3-b), b + min(a, 3-b)),
            (a + min(b, 4-a), b - min(b, 4-a))
        ]

        for state in states:
            if state not in visited:
                q.append(state)

water_jug()
```

IDLE Shell 3.13.3

File Edit Shell Debug Options Window Help

```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>

===== RESTART: C:\Users\ganes\AppData\Local\Programs\Python\Python313\ai.py ====
0 0
4 0
0 3
4 3
1 3
3 0
1 0
3 3
0 1
4 2
4 1
0 2
2 3
2 gallons obtained!
>>>
```